BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Iteration: FOR Loops

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does selection do?

2. What does the Program Counter hold?
   A) The next instruction's address
   B) The last calculation result
   C) All saved user files
   D) Data moving from memory

3. "Larger cache increases the number of installed CPU cores." Correct this. Explain why.

4. CPU A: 4 GHz, two cores, 2 MB cache. CPU B: 3 GHz, eight cores, 8 MB cache. Why is neither fastest for every task?

Today's objectives

Recognise repeated instructions and write counted loops. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

FOR loop
Repeats a block for each count in a specified inclusive range.

Counter
The loop variable changes automatically each iteration.

NEXT
NEXT marks the end of a FOR loop and triggers the next iteration.

Counted iteration
Useful when the number of repetitions is known in advance.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

FOR loop
Repeats a block for each count in a specified inclusive range. For example, consider a daily exercise counter.

Counter
The loop variable changes automatically each iteration. For example, consider a quiz warm-up.

NEXT
NEXT marks the end of a FOR loop and triggers the next iteration. For example, consider a ticket numbering.

Counted iteration
Useful when the number of repetitions is known in advance. For example, consider a row display.

I Do - Worked model

A daily exercise counter is being developed. Explain how for loop and counter are used. [4 marks]

- Identify where this particular program needs control or values.
- FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1].
- Counter [1]; The loop variable changes automatically each iteration. [1].
- Give one accurate point and a linked reason for each technique.

We Do - Guided practice

1. A quiz warm-up is being programmed. Explain why for loop matters. [2 marks]

2. For a ticket numbering, explain how counter is used. [2 marks]

3. A row display needs next and counted iteration. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Fill the loop end value so "Back up" prints three times.

```
for attempt = 1 to ...
print("Back up")
next attempt
```

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how counter and next can help a competition scoreboard. [4 marks]

Q4. Explain the roles of for loop, next and counted iteration when creating a maths drill. [6 marks]

Exit Ticket - True or False?

Repeats a block for each count in a specified inclusive range.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It chooses a branch using a condition. Selection can be nested.

2. A - The next instruction's address

3. A larger cache holds more often-used data and instructions. This can reduce slower memory access.

4. A has more cycles per second, which may help one task. B has more cores for parallel work and more cache to reduce slow memory access. Speed also depends on the task and CPU design.

I Do - Explain the model

Identify where this particular program needs control or values.

FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1].

Counter [1]; The loop variable changes automatically each iteration. [1].

Give one accurate point and a linked reason for each technique.

We Do - Model answers

1. FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1].

2. Counter [1]; The loop variable changes automatically each iteration. [1].

3. NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1]. Counted iteration [1]; Useful when the number of repetitions is known in advance. [1].

Code First - Answer

```
for attempt = 1 to 3
print("Back up")
next attempt
```

Expected output: Back up
Back up
Back up

You Do - Self-mark

Q2: FOR loop [1]; NEXT [1].

Q3: Counter [1]; The loop variable changes automatically each iteration. [1]. NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1].

Q4: FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1]. NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1]. Counted iteration [1]; Useful when the number of repetitions is known in advance. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.